ActInf Textbook Group ~ Cohort 3 ~ Meeting 9 (Chapter 4, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 9 (Chapter 4, part 2) | 56:06

All right, greetings. It's March 22nd, 2023. We're in Cohort 3 and Meeting 9 in our second discussion on Chapter 4 of the textbook.

So last time we went through some sections and we have also a lot of questions that have been addressed.

So before we go back to the text or to the questions, does anyone who's here want to just like give any thought or reflection?

Yeah, what are you still looking to...

reduce your uncertainty on?

Yeah, I'm just finding a lot of the mathematics.

It's just not explained in a way that disambiguates what a particular expression might be.

I've got certain uncertainties about some of the mathematical objects themselves.

And then particularly on the section on Taylor series, and I feel that I have a pretty good grasp on Taylor series.

That's feeling particularly opaque to me, which feels surprising.

Yeah, I'm finding it a slightly frustrating chapter, which sort of dives into the maths, but in some sense without enough detail for me to really deeply understand it.

But maybe that's just a matter of time.

Thank you.

Thank you.

Well, definitely when building on a sort of meat and potatoes mathematical concept, someone with your expertise should not be overly confused.

It should feel like you have already been on the on-ramp to understanding where this is going.

Because it's hard to imagine it would be clearer for someone who is less familiar with the foundational pieces.

So let's definitely keep that in mind when we move through.

Okay.

Anyone else want to just like give any overview thought on a section of four or anything else on how they're thinking about any aspect of it since last week?

Okay.

Okay.

Okay.

Then we will look at the questions.

And then continue to move through the text.

I'm just looking at the previous video to see where we ended.

Okay.

Okay.

Okay.

Okay.

Okay.

Any other comments?

Otherwise, we're going to go to 410 and continue or a little bit before.

Okay.

All right.

Let's pull back to 4-4 and just please raise your hand.

Or write in the chat with any thoughts or questions.

So just looking at the questions that were submitted in this cohort.

Many of them involve the equations.

4, 7, 8, 10, and so on.

So these are ones where like we can look at what was written for those where things were written or otherwise just like kind of given the constraints on what we can do in 50 minutes.

So just recognize these are like important areas for there to be asynchronous development on.

But this looks like a great answer.

Whomever has written this and it's like an exemplar of how to address it asynchronously.

Yes, Jonathan.

Yeah.

So this was my contribution here.

The explanation underneath.

I still have some questions about it, despite the fact that I think I now understand the sort of the vector aspect.

So I have created an Overleaf document that sort of explains all of this.

I'm still slightly confused by something quite simple here in the expected free energy, because there are these tau indices floating around and tau indicates, I think the time, the moment in time.

So for instance, s of p subscript pi tau is the probability of being in a given state at a particular time, given that you're undertaking some policy pi.

And so there's a tau index floating around, but it's still not quite clear to me in the calculation of the expected free energy what you do with that tau index.

Are you summing over all of these terms?

Are you integrating over them?

It is an expectation value over all of the different time steps.

So that was the one thing that was sort of least clear to me here.

Yes.

Great point.

Is it expected free energy as an expectation of snapshot free energies?

Is it expected free energy as the sum of the expected futures?

Or is it even the expectation at that last time step alone?

And the other ones are discarded.

Good points.

Good points.

In figure 4-3, we have the discrete time, partially observable Markov decision process, POMDP, and the continuous time cousin.

In the discrete time case, past, present, and future time steps, hidden states are explicitly predicted.

And action sequences are explicitly intervening in how hidden states change their time.

In contrast, in the continuous time model, rather than predicting or even explicitly mentioning past, present, and future, we see like x, x prime, x double prime.

So the value of x, and then the higher derivatives, also known as the generalized coordinates of motion.

And so this amounts to making a Taylor series expansion, which is continuous, around the current time step.

Again, there are two different ways of dealing with temporal depth and the way that action selection relates to temporal depth.

And they have some, in principle and in practices, strengths and weaknesses in different settings.

focus on the discrete time model. The discrete time model has been especially applied to categorical cognitive decision making, whereas the archetypal case for the continuous time generative model is continuous perception and action of like a motor unit. And they can play well together as we'll see in chapter 8 with hybrid models. We could go slower fast on equations, but we'll just try to like move through the whole thing. But many, many of these, the work to be done is making sure that we have high quality descriptions verbally, which we do for many, due to many people's great work, but making sure that we have verbal descriptions in the active inference ontology that are faithful to the symbols. And then where people have questions about what does this mean? Why is that that way? What if it were this way? What are the implications here? Those are all great questions and they can all be included in the notes section for a given equation or in the more general notes table. But for many of the equations, including some of the super challenging ones, we have things written out, which is very good. But just kind of at the skim level, the cat is a categorical distribution. So in this section, we're in a discrete setting. So like the light is on or off. The continuous setting would be it's a continuous variable between zero and one. But in the categorical setting, we have categorical distributions. And so we have certain ways of summing across them, for example, instead of taking an integral over a continuous distribution.

Action selection, pi, policy selection,

is going to be guided by which policies are most likely. Most self-consistent, most self-evident, most probable, lowest expected free energy. We're not proposing a reward or utility function and then evaluating policies based upon their expected reward.

We're evaluating policies based upon their self-consistency. And in fact, the most probable ones are the ones that lead to the lowest expected free energy.

So, more equations. G is expected free energy. And even just looking at which variables go into this, the lowest, we see a probability distribution over policies that is defined through an expected free energy calculation. And this G of pi, it's a function of policy, and it is going to have this expected free energy construction. And we've seen expected free energy before in equation 2.6,

where the part that's an expectation that only involves preferences and observations is pragmatic value.

So, natural log of a distribution expectation of a log of something only involving observations and preferences.

Here, we have y tilde. Here, we have o tilde. Y is sometimes used for observations more in like a linear

regression setting. O is like observations, but they're identical. And then the other component is a

and so it's pragmatic value because that is the alignment between the observations and the preferences

or expectations. The other component in this decomposition of expected free energy is a KL divergence, expected KL

of the same time to be a KL divergence between two variational distributions, two distributions that we control, Q.

They're both X, hidden states through time, X tilde, conditioned upon, and then they're very similar.

Here, we only have pi. Here, we have pi and y. So, this is the divergence

with respect to whether adding observations updates are prior. If these two queues are the same,

that's equivalent to saying there's no space between the distributions where we've parameterized

X with or without getting any observations. In other words, the observations were valueless

if the divergence here is zero. Whereas, if the observations are really impactful on moving the

distribution, then this KL divergence is going to be big. Divergences are always non-zero.

And then, there's a negative here. So, the more information gain, the bigger the KL divergence,

the more that the observations matter, the more that this first term is going to go negative.

And so, that specific policy will have a higher epistemic value.

Ali?

Ali?

Yeah, I just remembered a clarification made by Ryan Smith in one of his lectures about

two concepts of or rather two notions of time used in active inference literature,

one of which is denoted by tau and the other one is denoted by T. So, tau is used for the time

about which we have a belief. And the examples he gives is, for instance, I believe I am now in my

car. So, that would be S of tau. But I believe I was in my kitchen 10 minutes ago would be S

tau minus 1. Or I believe I will be at work in 20 minutes or something would be S tau plus 1.
So, but on the other hand, we have T, which refers to the time at which a new observation is given.
So, for instance, as an example, and after turning on a light, I now, in other words, S of tau, believe that I was in the kitchen for the last five minutes, right? So, we can update our belief about all all tau's or S of tau's with each new observation at each T's or at each given observation times. So, briefly, one is the belief time and the other one is the observation time. So, tau is used for belief times or the time about which we have belief and T is used for observation times.
Thanks. That's very helpful. Is it accurate to say that T is like the click of our simulation that's actually handing the new observations in at a given moment, T after T, but then, especially in sophisticated active inference where we might be considering like the past, present, and future at every T, we are reevaluating tau's at each T. Yeah, I think we can reframe that in this way as well.
Awesome. So, here we reach, yes, potentially with some bumps and jumps, the expected free energy for a policy in terms of the pragmatic value, alignment of preferences with observations, and the expected information gain, the epistemic value, in terms of how much adding in observations updates our variational distribution about hidden states conditioned upon that policy.
Here is more focus on how we get this KL divergence.
So, here we see that KL divergence, and there's just a few different expansions and relationships around it. Jonathan?
Yeah, so just zooming into that equation there, there's an expectation over Q of O given pi.
Jonathan? And I couldn't figure out how that's calculated because I understand how one can get Q of S given O and pi, but it seems that to calculate Q of O given pi, one needs to marginalize or sum over the states. And I thought that that was the whole reason for introducing Q in the first place, was because we can't sum over all external states.
The subscript notation may be
used in an unconventional way, but I agree with you on that.
Here, in the inner Q, we have it about S, conditioned upon observations and policies. This doesn't seem contentious. Yeah, that's fine.
Yeah. But the subscript in 4.8, and I think that was probably one of your questions.
Yeah.
Yeah.
Yeah.
Yeah, that's the one. Sorry, with the terrible lack of...
Yeah, yeah, yeah. Sorry.
Yeah. Yeah. Yeah. We should look at the MATLAB code for wherever that comes into play to see how it actually is done. Or is it a separate variational quality?
Yeah. Yeah.
Yeah. Like, this is the... just the inner value is the KL divergence of how much observations update...
...add in to your changing beliefs about S. But we're taking an expectation of that over expected

observation sequences, which then begs the question of needing to know the distribution of S

S that would generate those probabilistic envelopes of O.

Exactly.

Yeah.

Well, one answer is these are the analytical ideals for which there are strong analytical guarantees,

and then any given computational implementation is going to use different methods to implement this

just procedurally.

But that doesn't make the notation clear.

And I think these are some of the kinds of artifacts, hopefully cohort after cohort,

that we can craft. Like a lookup table.

And we've done this to a limited extent, just starting it.

But having a lookup tables for all the different shapes and squiggles.

But continuing on, so I hope that we can at least see the rest of it.

So, we'll recall from equation 2.5 is a functional of Q, the distribution we control, and Y, the incoming data.

So, variational free energy is like the real-time unfolding model fit.

And this is a quantity that's widely used in statistics.

Expected free energy uses similar structure and positioning,

but it explicitly brings in two pieces that aren't present in variational free energy.

And those are observations which haven't happened yet, because we need to talk about how informative we expect observations that haven't occurred yet to be.

Information gain, epistemic value.

Obviously, variational free energy is only concerned with the actual data point being acquired at the moment.

And then secondly, the fact that we can take different policy choices.

And policy choices end up conditioning, for example, what pragmatic and epistemic value we achieve or expect to achieve.

So, free energy can have a variational free energy formulation, real-time unfolding model adequacy.

Or, projecting free energy into the expected future, we need to invoke observations that haven't happened yet,

and our agency in selecting alternative policies, which then, of course, bear on the epistemic and pragmatic value we accomplish.

Here, in 4.9, there are more decompositions of expected free energy.

That help see previously recognized dialectics or decompositions or trade-offs as actually decompositions of the unified imperative expected free energy.

So, epistemic and pragmatic value decomposition and an ambiguity plus risk.

And one can kind of walk through and see what those mean or look at the natural language.

We can rewrite 4.7 in linear algebraic form.

Here's 4.7.

Here is, for the categorical distribution, we can sum over policies about the policies and the hidden states through time.

More details and calculations of variational free energy.

Factorization and the mean field approximation.

These are some mathematical technicalities, but they are simplifying assumptions or ways to approach this in the general case.

This factorization is one of many possibilities in variational inference.

And as we talked about last time, variational inference is just one way to do these calculations.

And represents the simplest option.

And in practice, it's nuanced slightly, and there's like other approximations, and it's a whole thing.

More information on the letters that we know and love.

A. The Tale of Two Densities.

B. How Hidden States Change Through Time.

C. Prior Beliefs About Observations, Which Are Preferences.

C. Prior, and the Simple Prior on the Initial State to Kick the Hidden State Chain Off.

Little bit of Policy and Belief Updating, a few more technical details.

Into Active Inference and Continuous Time.

B. Box 4-1 on Markov Blankets.

A Markov Blanket for a given variable.

With just those words, the map territory question is addressed.

Markov Blankets are about variables.

Comprises a subset of those that interact with it.

If we know everything about the subset,

knowledge of anything outside the subset does not increase our knowledge of the variable of interest.

There are several

Bayesian message passing schemes

expounded on a lot further

in other work with

Friston, Parr, DeVries, et al.

Probably not

the most relevant to delve into

right now,

but the important takeaway

is that

by using a Bayes graph,

so by making a commitment

to

the figure 4.3

Lego set,

if we choose to have

nodes be random variables
and edges reflect relationships amongst variables
in a certain way,
then
we gain access
to
message passing
techniques
that make
even large
Bayesian graphs
with different parts of the graph
updating at different speeds
and so on.
There are software packages
that are able to,
in a defined
computational complexity setting,
run inference
on those Bayes graphs.
So yes,
bigger models,
all things being equal,
take more time
and more computational resources to run.
However,
there are some
tremendously scalable
!
There are some
high-performance ways
to computationally implement
computation
on these graphs.
And they have
interesting analytical representations
and
they have

a growing
set
of toolkits
like
reactive
message
passing
dot JL
by the Bias Lab
with Bert DeVries
and
his lab.
This is like
really cool
current work
that's not even
outside of their
lab and colleagues
not heavily
used
in
even
active
inference
publications.
But
they have
incredible
recent
work.
And you can see
how they're using
variational free energy
in these
advanced
settings.
Pulling back
to consider

message passing
qualitatively
in the chapter
whereas it was
introduced
in this
Markov blanket
box
message
passing
though
I'd be
totally
looking
oh
thank you
I will
yeah
there's
the Bias Lab
site
and I'll
add it
to
the
figure
four
the
reactive
message
passing
is like
one of
their
new
pieces
where
let's say

you had
two sensors
and one
of them
was being
updated
every second
and one
was being
updated
every hour
well if you
only could
turn the
clock
if you
you might
get into a
situation
where you
were either
say well
should I
update my
model every
second
and then
over sample
the slow
variable
or should I
sample my
model every
hour
and under
sample the
fast variable
and reactive

message passing
navigates that
by actually
having nodes
update
on demand
so you can
have one
node that's
just getting
pinged every
second
and another
node that's
getting updated
on a different
time scale
and the
computations on
demand
so that
allows you to
have like
lifelong
online learning
for base
graphs
flexibly
with different
rates of
sensors
and so
those are the
kinds of
toolkits
like
these models
are not just

being proposed
to promote
theoretical unity
in the
cybernetic
sciences
also
there are
powerful
software
toolkits
that
using
these
kinds of
architectures
enable
so epistemic
value
pragmatic
value
message
passing
schemes
if anyone
has any
takeaways from
this image
please feel
free because
certain things
are like
omitted
and also
it uses
like a lot
of squiggles
and ease

and different
variables
but
here we
have a
representation
of
a
hierarchical
model
so you can tell
it's a hierarchical
model because you
have like i
and you have this
kind of cluster of
four
and you have i
plus one
and you have
errors getting
passed up
and hidden
state inferences
getting passed
down
and so
this would be
like the sort
of timeless
representation of
that modeling
hierarchy
hierarchical
base
hierarchical
predictive
processing

architecture
tilde
through time
so we're just
describing
timelessly
here is the
hierarchical
bayesian
model
this
can be
rendered
or unpacked
through time
now we see
that tau
minus one
tau
tau
plus one
as a
function
of time
so instead
of just
seeing it
as like
o tilde
like a
sequence
and seeing
this as
like a
skyscraper
this can
be unrolled
in a

procedural
way
using
message
passing
techniques
kate
are the
rules for
message
passing
on a
convoluted
bayesian
network
known
yes
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paper
that i
have just
put in
the chat
here we
see
figure
4.3
discrete
time
again
and here
is a
alternative
representation
of this
base graph
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is known

as a
forney
factor
graph
and so
in this
paper
they show
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any
base
graph
can be
represented
as a
forney
factor
graph
and any
forney
factor
graph
has a
defined
procedural
message
passing
algorithm
so
actually
a quite
incredible
synthesis
even in
this
2017
paper
because it

means if
you can
build with
these
legos
with this
semantically
interpretable
setting
then
there will
always be
a procedural
technique
to calculate
it
no matter
how
convoluted
this graph
is
which is
very important
because you
could make
you could just
add linear
model layers
but then
you would
start to
lose
analytical
guarantees
very quickly
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4.4
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focuses more
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continuous
time
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4.15
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line
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function
true
underlying
function
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our
first
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quadratic
expansion
drawing
an upside
down
parabola
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point
expanding
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discussions
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above

into
hierarchical
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generalized
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coding
schemes
and a
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an interesting
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details
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message
passing
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coordinates

and
Laplace
approximation
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of
challenges
and
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because
this is
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active
inference
chapter
one
overview
chapter
two
low
road
this
is
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to
do
it
chapter
three
high
road
this
is
why
we're
going
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four
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models
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inference
figure
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generative
models
and
then
there's
a lot
of
technical
details
so
anyone
have
any
just
overall
thoughts
or
questions
on
four
Jonathan
and
then
anyone

else
yeah
I
think
just
to
get
more
of
a
sense
of
it
I'd
still
be
interested
to
know
more
about
the
tau
in
g
if
anyone
has
any
insight
into
that
when
you
calculate
g
yeah
does

g
still
have
a
tau
index
or
is
something
else
happening
to
it
yes
good
good
that
it's
noted

If
anyone
has
a
link
or
thought
go
for
it
otherwise
that
is
definitely
something
that
we
will

need
to
revisit
and
improve
for
future
cohorts
Kate
just
a
very
general
comment
like
this
was
way
over
my
head
overall
but
I
was
just
wondering
if
in
your
experience
the
textbook
that
you
are
kind
of

working
through
with
I
mispronounce
his name
name
Joshi
about
the fundamentals
of active
inference
do you
think there
will be
some help
in any
of these
areas?
Ali
what's your
first thought
on that?
This is in
reference to
Sanjeev
Nemjoshi's
fundamentals
textbook that
we are
working through
and evaluating
during 2023
so Ali
what would you
say to
Kate's
question?

yeah
in my
opinion
I think
it is
way
way
more
helpful
in
understanding
the
mathematical
details
and
also
in
fact
the
conceptual
intricacies
that
have been
somehow
glossed
over
in this
textbook
but
on the
other
hand
it is
estimated
to be
about
600 to
900 pages

long
so
yeah
it is
much more
expanded
than this
textbook
and it
goes into
a very
helpful
detail
expanding
upon
these
mathematical
derivations
but
one point
I wanted
to make
is that
Sanjeev's
textbook
is not
strictly
geared
toward
somewhat
mathematical
derivation
of these
equations
but
its
main
objective

is
to use
it
as a
practical
tool
in order
to
model
these
kinds
of
agents
and
in fact
he also
provides
some
helpful
Jupyter
notebooks
in that
regard
but
yes
I think
it can
be
really
helpful
in bridging
the gap
between
this
textbook
and the
practical
side

of
doing
things
yeah
thanks
so
the book
is not
available
yet
it's
not
even
finished
in
writing
if
anybody
wants
to be
added
to
view
chapters
and the
code
which are
all in
python
which also
increases the
accessibility
relative to
the matlab
of this
part
all
these

chapters
in a
way
that
has not
appeared
in our
space
before
are
tackling
these
statistical
questions
so it
is still
like very
much an
accelerated
statistical
learning
curve
but
as Ali
mentioned
it's less
focused on
like well
here's how
you get
from
here to
here as
just
as
an
analytical
derivation

consideration
and it's
a lot
more like
we're
going to
go step
by step
with
building
stochastic
simulations
and
so this
is going
to be a
great
compliment
and
fields
have many
textbooks
so there's
not just
one
right
approach
but
it's been
really encouraging
to see
his work
so if
anybody
wants
like
even
just

access
to
the
coda
to
review
the
chapters
please
feel free
to email
us
and
hopefully
in
the
coming
year
we'll
have
more
details
on
when
it'll
be
available
let
us
look
ahead
quickly
to
chapter
five
chapter
five
is the

last
chapter
in the
first
half
of the
textbook
which
is
the
epistemic
half
of the
textbook
so
congratulations
to those
who have
continued
on
especially
through
that
second
half
of
chapter
four
which
is
definitely
like
intense
to a
comical
degree
and so
carefree

and whimsical
in its
writing
style
which
which
has
its own
kind of
comedy
but we're
going to
have
two weeks
on
chapter
five
which
is
going
to
be
a
great
return
to
reality
and
then
the
final
meeting
will be
on
feedback
project
ideas
Terry

I think
we need
to
approach
to the
maths
pitched
at a
seven
year
old
yeah
many
angles
on
that
one
is
in
the
math
learning
group
we've
explored
these
kinds
of
questions
and
another
approach
is
a
non
mathematics
approach
which

several
people
have
investigated
how
important
will it
be
if we
didn't
grasp
all the
details
of
chapter
four
when
we
get
to
chapter
five
irrelevant
chapter
five
picks up
on a
very
different
note
remember
that
chapter
four
they
even
said
you

could
skip
said
if
you
don't
want
to
know
the
technical
details
you
can
skip
chapter
four
which
was
again
like
wait
but
we
just
took
the
low
road
and
the
high
road
to
our
city
and
then

they're
like
you
don't
have
to
go
on
the
tour
if
you
don't
want
to
chapter
five
message
passing
and
neurobiology
Bayesian
message
passing
comes
back
so
again
that
box
that
we
looked
at
earlier
the
idea
that

every
Bayes
graph
is
sparse
not
everything
depends
on
everything
because
of the
sparsity
of the
Bayes
graph
especially
there
are
some
computational
procedures
that
are
extremely
effective
and
interestingly
real
biological
systems
also
have
sparse
connectivity
whether
it's
the

neural
architecture
in the
brain
or
whether
it's
the
interaction
patterns
in an
ant
colony
in
distributed
systems
connections
are
sparse
they
begin
at
the
micro
anatomical
level
with
looking
at
neurons
and
their
connections
they
focus
on
one
of

the
most
well
studied
neural
tissues
which
is
the
mammalian
cortex
and
the
mammalian
cortex
has
what's
called
a
columnar
architecture
which
is
these
six
histologically
distinct
layers
of
tissue
so
they
look
different
under
a
microscope
and

the
connectivity
patterns
in terms
of what
actually
sends
axons
where
are
also
essentially
completely
characterized
and so
in here
we see
the
tantalizingly
realist
alignment
of the
histological
columnar
micro
architecture
with
bayesian
graphs
that
have
similar
structure
in
figure
5.2
they're
going to

generalize
and kind
of
coarse
grain
the
columnar
architecture
and now
look at
how
different
columns
communicate
laterally
so we
have
hierarchical
communication
within
a
columnar
unit
and
connections
across
units
this is
kind
of
like
Numenta
thousand
brain
hypothesis
like
a lot
of

work
that
is
using
the
neocortical
architecture
in search
of intelligent
machines
so that's
the
neocortex
scenario
we're
going to
switch
systems
to the
mammalian
motor
cord
here
we have
incoming
proprioceptive
data
y
and
descending
predictions
those
predictions
and
observations
are
juxtaposed
to form

an
error
signal
which
gets
passed
forward
so this
is a
continuous
time
setting
and
here
we can
see
action
as
resolving
discrepancies
between
target
set
points
and
proprioceptive
input
now we
switch to
another
mammalian
neuroanatomical
system
which is
subcortical
and the
basal ganglia
in particular

which is a
dopaminergic
region
this region
plays a role
in policy
selection
check out
the
oliver
sacks
book
or
robin
williams
movie
awakenings
it's
really
interesting
and funny
here we
see two
different
ways that
the
dopaminergic
tone
tilts

the
balance
towards
either
action
selection
which
passes

habits
forward
or
that
kind
of
type
one
indirect
policy
selection
pathway
can tip
towards
a type
two
in a
kahneman
sense
deliberative
free
energy
updated
policy
posterior
but we'll
talk more
about it
they talk
about
different
neurotransmitters
in
active
they talk
about
how
continuous

and
discrete
models
can come
together
and then
they end
the chapter
with bringing
together
those three
systems
that were
previously
described
the
columnar
architecture
in the
neocortex
the
basal
ganglia
dopaminergic
decision-making
architectures
and
the
spinal
cord
reflex
proprioceptive
continuous
time
model
and
they
show

how
those
kinds
of
models
can
be
integrated
to
address
some
of
our
favorite
topics
so
it's
a shorter
chapter
it doesn't
have
equations
like the
previous
ones
do
so
again
congratulations
to everybody
who made
it through
the more
thick of
it
chapter
four
and

chapter
five
we're
going to
have
a lot
of
space
for
conversations
about
neurophysiology
and
the body
and
it's
just a
good
close
to the
first
half
of the
textbook
so
thank you
all
for
joining
see you
next week
five